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Motif Shapes and State-Class Catalogue

Simon Frost 2026-05-14

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Introduction

The motif/subgraph-closure framework of [Keeling, House, Cooper & Pellis \(2016\)](#) tracks, for each connected m -vertex induced subgraph *shape* H of the host network and each canonical state assignment σ of H 's vertices, the count $E_{H,\sigma}(t)$ of such induced copies in the current state. Two embeddings are identified if one is sent to the other by a graph automorphism of H ; this reduces the state space from $|Q|^m \times |\text{Emb}(H, G)|$ to a manageable set of orbit representatives.

Closure enters at the $(m+1)$ -th order: every $(m+1)$ -vertex induced amplitude is expressed as a Kirkwood-type ratio of tracked m -vertex and lower-order quantities. For the SIS model on a k -regular host this gives an autonomous ODE system whose variable count grows with (k, m) but remains dramatically smaller than the full 2^N master equation.

This vignette enumerates every shape and every SIS state class for all supported (k, m) pairs: $k = 2$ with $2 \leq m \leq 6$, and $k = 3$ with $m \in \{2, 3, 4\}$. It is generated programmatically from the package's own `enumerate_shapes` and `enumerate_state_classes` functions so the tables are guaranteed to match the running implementation.

Setup

```
using NodeBasedModels
```

Counting convention

Variables count unordered induced embeddings of a shape H partitioned by canonical state class under $\text{Aut}(H)$. Conversion to the “directed-pair” or “labelled” counts used in the standard Keeling pairwise convention multiplies by the orbit size $|\text{orb}(\sigma)|$:

$$L_\sigma = E_{\text{canon}(\sigma)} \cdot |\text{orb}(\sigma)|,$$

where $|\text{orb}(\sigma)| = |\text{Aut}(H)|/|\text{Stab}(\sigma)|$.

For the P_2 shape ($|\text{Aut}| = 2$):

Canonical class	Orbit size	Relation to Keeling pairwise
$[SS]$	1	$[SS]_{\text{pw}} = 2E_{SS}$
$[IS]$	2	$[SI]_{\text{pw}} = E_{IS}$
$[II]$	1	$[II]_{\text{pw}} = 2E_{II}$

Catalogue generation

```
function fmt_state(st)
    join(string.(st), ",")
end

function shape_ascii(name::Symbol)
    d = Dict{
        :singleton => "●",
        :P2        => "●—●",
        :P3        => "●—●—●",
    }
```

```

:C3      ⇒ "●●\n  L●",
:P4      ⇒ "●●●●",
:P5      ⇒ "●●●●●",
:P6      ⇒ "●●●●●●",
:K13     ⇒ "  ●\n●●●\n  ●",
:paw     ⇒ "●●\n | \n ●●",
:C4      ⇒ "●●\n | | \n ●●",
:K4me    ⇒ "●●\n | \n ●●",
:K4      ⇒ "●●\n |X| \n ●●",
)
get(d, name, string(name))
end

for (k, m) in [(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(3,3),(3,4)]
  closure = MotifClosure(k, m)
  shapes = NodeBasedModels.enumerate_shapes(closure)
  println("=" ^ 60)
  println("(k=$k, m=$m) - $(length(shapes)) shape(s)")
  println("=" ^ 60)
  total_vars = 0
  for sh in shapes
    classes = NodeBasedModels.enumerate_state_classes(sh, [:S, :I])
    n_vars = length(classes)
    total_vars += n_vars
    println()
    println("  Shape :$(sh.name)  |V|=$(sh.n_nodes)  |E|=$(length(sh.edges))",
           "  |Aut|=$(length(sh.automorphisms))  state-classes=$n_vars")
    if !isempty(sh.edges)
      println("  Edges: ", join(["($a,$b)" for (a,b) in sh.edges], " "))
    end
    println("
    |-----|")
    println("    Canonical state      | Orbit size |")
    println("    |-----|")
    for (st, osz) in sort(classes; by=x→x[1])
      println("    | [$(rpad(fmt_state(st),19))] | $(lpad(osz,10)) |")
    end
    println("    |-----|")
  end
  println()
end
println()
println("  Total variables for (k=$k, m=$m): $total_vars")
end

```

(k=2, m=2) – 2 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I	1
[S	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3

Edges: (1,2)

Canonical state	Orbit size
[I,I	1
[I,S	2
[S,S	1

Total variables for (k=2, m=2): 5

(k=2, m=3) – 3 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I	1
[S	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3

Edges: (1,2)

Canonical state	Orbit size

[I,I	]	1
[I,S	]	2
[S,S	]	1

Shape :P3 |V|=3 |E|=2 |Aut|=2 state-classes=6
Edges: (1,2) (2,3)

Canonical state	Orbit size
[I,I,I	1
[I,I,S	2
[I,S,I	1
[I,S,S	2
[S,I,S	1
[S,S,S	1

Total variables for (k=2, m=3): 11

(k=2, m=4) – 3 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I	1
[S	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3
Edges: (1,2)

Canonical state	Orbit size
[I,I	1
[I,S	2
[S,S	1

--	--

Shape :P4 |V|=4 |E|=3 |Aut|=2 state-classes=10
Edges: (1,2) (2,3) (3,4)

Canonical state	Orbit size
[I,I,I,I]	1
[I,I,I,S]	2
[I,I,S,I]	2
[I,I,S,S]	2
[I,S,I,S]	2
[I,S,S,I]	1
[I,S,S,S]	2
[S,I,I,S]	1
[S,I,S,S]	2
[S,S,S,S]	1

Total variables for (k=2, m=4): 15

(k=2, m=5) - 3 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I]	1
[S]	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3
Edges: (1,2)

Canonical state	Orbit size
[I,I]	1
[I,S]	2

[S,S]	1
-------	---

Shape :P5 |V|=5 |E|=4 |Aut|=2 state-classes=20

Edges: (1,2) (2,3) (3,4) (4,5)

Canonical state	Orbit size
[I,I,I,I,I]	1
[I,I,I,I,S]	2
[I,I,I,S,I]	2
[I,I,I,S,S]	2
[I,I,S,I,I]	1
[I,I,S,I,S]	2
[I,I,S,S,I]	2
[I,I,S,S,S]	2
[I,S,I,I,S]	2
[I,S,I,S,I]	1
[I,S,I,S,S]	2
[I,S,S,I,S]	2
[I,S,S,S,I]	1
[I,S,S,S,S]	2
[S,I,I,I,S]	1
[S,I,I,S,S]	2
[S,I,S,I,S]	1
[S,I,S,S,S]	2
[S,S,I,S,S]	1
[S,S,S,S,S]	1

Total variables for (k=2, m=5): 25

(k=2, m=6) – 3 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
-----------------	------------

[I	]	1
[S	]	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3
Edges: (1,2)

Canonical state	Orbit size
[I,I	1
[I,S	2
[S,S	1

Shape :P6 |V|=6 |E|=5 |Aut|=2 state-classes=36
Edges: (1,2) (2,3) (3,4) (4,5) (5,6)

Canonical state	Orbit size
[I,I,I,I,I,I	1
[I,I,I,I,I,S	2
[I,I,I,I,S,I	2
[I,I,I,I,S,S	2
[I,I,I,S,I,I	2
[I,I,I,S,I,S	2
[I,I,I,S,S,I	2
[I,I,I,S,S,S	2
[I,I,S,I,I,S	2
[I,I,S,I,S,I	2
[I,I,S,I,S,S	2
[I,I,S,S,I,I	1
[I,I,S,S,I,S	2
[I,I,S,S,S,I	2
[I,I,S,S,S,S	2
[I,S,I,I,I,S	2
[I,S,I,I,S,I	1
[I,S,I,I,S,S	2

[I,S,I,S,I,S]	]	2
[I,S,I,S,S,I]	]	2
[I,S,I,S,S,S]	]	2
[I,S,S,I,I,S]	]	2
[I,S,S,I,S,S]	]	2
[I,S,S,S,I,S]	]	2
[I,S,S,S,S,I]	]	1
[I,S,S,S,S,S]	]	2
[S,I,I,I,I,S]	]	1
[S,I,I,I,S,S]	]	2
[S,I,I,S,I,S]	]	2
[S,I,I,S,S,S]	]	2
[S,I,S,I,S,S]	]	2
[S,I,S,S,I,S]	]	1
[S,I,S,S,S,S]	]	2
[S,S,I,I,S,S]	]	1
[S,S,I,S,S,S]	]	2
[S,S,S,S,S,S]	]	1

Total variables for (k=2, m=6): 41

(k=3, m=2) – 2 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I]	1
[S]	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3

Edges: (1,2)

Canonical state	Orbit size

[I,I	]	1
[I,S	]	2
[S,S	]	1

Total variables for (k=3, m=2): 5

(k=3, m=3) – 4 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I	1
[S	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3
Edges: (1,2)

Canonical state	Orbit size
[I,I	1
[I,S	2
[S,S	1

Shape :P3 |V|=3 |E|=2 |Aut|=2 state-classes=6
Edges: (1,2) (2,3)

Canonical state	Orbit size
[I,I,I	1
[I,I,S	2
[I,S,I	1
[I,S,S	2
[S,I,S	1
[S,S,S	1

--	--

Shape :C3 |V|=3 |E|=3 |Aut|=6 state-classes=4
Edges: (1,2) (2,3) (1,3)

Canonical state	Orbit size
[I,I,I]	1
[I,I,S]	3
[I,S,S]	3
[S,S,S]	1

Total variables for (k=3, m=3): 15

(k=3, m=4) – 10 shape(s)

Shape :singleton |V|=1 |E|=0 |Aut|=1 state-classes=2

Canonical state	Orbit size
[I]	1
[S]	1

Shape :P2 |V|=2 |E|=1 |Aut|=2 state-classes=3
Edges: (1,2)

Canonical state	Orbit size
[I,I]	1
[I,S]	2
[S,S]	1

Shape :P3 |V|=3 |E|=2 |Aut|=2 state-classes=6
Edges: (1,2) (2,3)

--	--

Canonical state	Orbit size
[I,I,I]	1
[I,I,S]	2
[I,S,I]	1
[I,S,S]	2
[S,I,S]	1
[S,S,S]	1

Shape :C3 |V|=3 |E|=3 |Aut|=6 state-classes=4

Edges: (1,2) (2,3) (1,3)

Canonical state	Orbit size
[I,I,I]	1
[I,I,S]	3
[I,S,S]	3
[S,S,S]	1

Shape :P4 |V|=4 |E|=3 |Aut|=2 state-classes=10

Edges: (1,2) (2,3) (3,4)

Canonical state	Orbit size
[I,I,I,I]	1
[I,I,I,S]	2
[I,I,S,I]	2
[I,I,S,S]	2
[I,S,I,S]	2
[I,S,S,I]	1
[I,S,S,S]	2
[S,I,I,S]	1
[S,I,S,S]	2
[S,S,S,S]	1

Shape :K13 |V|=4 |E|=3 |Aut|=6 state-classes=8
Edges: (1,2) (1,3) (1,4)

Canonical state	Orbit size
[I,I,I,I]	1
[I,I,I,S]	3
[I,I,S,S]	3
[I,S,S,S]	1
[S,I,I,I]	1
[S,I,I,S]	3
[S,I,S,S]	3
[S,S,S,S]	1

Shape :paw |V|=4 |E|=4 |Aut|=2 state-classes=12
Edges: (1,2) (1,3) (2,3) (1,4)

Canonical state	Orbit size
[I,I,I,I]	1
[I,I,I,S]	1
[I,I,S,I]	2
[I,I,S,S]	2
[I,S,S,I]	1
[I,S,S,S]	1
[S,I,I,I]	1
[S,I,I,S]	1
[S,I,S,I]	2
[S,I,S,S]	2
[S,S,S,I]	1
[S,S,S,S]	1

Shape :C4 |V|=4 |E|=4 |Aut|=8 state-classes=6
Edges: (1,2) (2,3) (3,4) (4,1)

Canonical state	Orbit size
-----------------	------------

[I,I,I,I	]	1
[I,I,I,S	]	4
[I,I,S,S	]	4
[I,S,I,S	]	2
[I,S,S,S	]	4
[S,S,S,S	]	1

Shape :K4me |V|=4 |E|=5 |Aut|=4 state-classes=9

Edges: (1,2) (2,3) (3,4) (4,1) (1,3)

Canonical state	Orbit size
[I,I,I,I]	1
[I,I,I,S]	2
[I,I,S,I]	2
[I,I,S,S]	4
[I,S,I,S]	1
[I,S,S,S]	2
[S,I,S,I]	1
[S,I,S,S]	2
[S,S,S,S]	1

Shape :K4 |V|=4 |E|=6 |Aut|=24 state-classes=5

Edges: (1,2) (1,3) (1,4) (2,3) (2,4) (3,4)

Canonical state	Orbit size
[I,I,I,I]	1
[I,I,I,S]	4
[I,I,S,S]	6
[I,S,S,S]	4
[S,S,S,S]	1

Total variables for (k=3, m=4): 65

Variable-count summary

```
println("| (k,m) | Shapes tracked | State classes | Variables |")
println("|-----|-----|-----|-----|")
for (k, m) in [(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(3,3),(3,4)]
    closure = MotifClosure(k, m)
    shapes = NodeBasedModels.enumerate_shapes(closure)
    names = join([":${sh.name}" for sh in shapes], " + ")
    nvars = sum(length(NodeBasedModels.enumerate_state_classes(sh, [:S,:I]))
                for sh in shapes)
    nclass = nvars
    println("| (k=$k,m=$m) | ${rpad(names,38)} | ${lpad(nclass,12)} | ${lpad(nvars,9)} |")
end
```

(k,m)	Shapes tracked	State classes	Variables
-----	-----	-----	-----
(k=2,m=2)	:singleton + :P2	5	5
(k=2,m=3)	:singleton + :P2 + :P3	11	11
(k=2,m=4)	:singleton + :P2 + :P4	15	15
(k=2,m=5)	:singleton + :P2 + :P5	25	25
(k=2,m=6)	:singleton + :P2 + :P6	41	41
(k=3,m=2)	:singleton + :P2	5	5
(k=3,m=3)	:singleton + :P2 + :P3 + :C3	15	15
(k=3,m=4)	:singleton + :P2 + :P3 + :C3 + :P4 + :K13 + :paw + :C4 + :K4me + :K4		

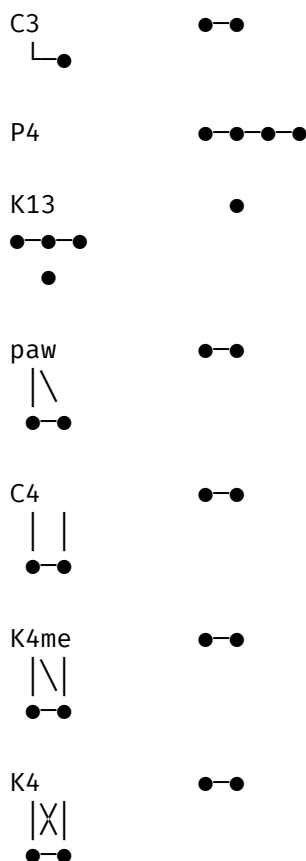
Shape diagrams

```
all_names = [:singleton, :P2, :P3, :C3, :P4, :K13, :paw, :C4, :K4me, :K4]
for nm in all_names
    println("${rpad(string(nm),10)}  ${(shape_ascii(nm))}")
    println()
end
```

singleton ●

P2 ●—●

P3 ●—●—●



Notes on the Lean-certified obstruction at (k=3, m=4)

The (k=3, m=4) implementation is intentionally documented as a boundary case, not as a guaranteed monotone refinement of m=3. Lean theorem T3b in `EdgeBasedModels.jl/proofs/EBCMCategory/Marginali` certifies that the current order-4 Kirkwood RHS need not marginalise to a better order-3 RHS; theorem T7 gives the corresponding small-time trajectory lower bound. In practical terms, simply adding the 4-vertex variables does not force the projected dynamics to improve the 3-vertex closure on the random 3-regular benchmark.

This does not prove that every possible (k=3, m=5) or higher closure is impossible. It says that further k=3 extensions require new analysis (for example, a non-Kirkwood or constrained-consistency closure) rather than a mechanical “more motifs is always better” interpretation.

The obstruction is a categorical fact about the shape of the *marginalisation functor* on the poset of induced subgraphs, not a numerical accident. See Vignette 11 for the live numeric witness and the formal statement.

References

- Keeling M.J., House T., Cooper A.J., Pellis L. (2016). Systematic Approximations to Susceptible-Infectious-Susceptible Dynamics on Networks. *PLoS Comput. Biol.* 12(9): e1005296. <https://doi.org/10.1371/journal.pcbi.1005296>
- Sharkey K.J. (2011). Deterministic epidemic models on contact networks: Correlations and unbiological terms. *Theor. Pop. Biol.* 79(4):115–129.
- Kirkwood J.G. (1935). Statistical mechanics of fluid mixtures. *J. Chem. Phys.* 3:300–313.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use `NetworkOutbreaks.jl`'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette `01_sir_on_graphs` for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.